

| Tool Name                       |                                                                             |                                   |                                   | Scaphandre                                 | Keple ≥ 0.10                                                                                                  | Kepler ≤ v0.9                                                                                                    | Green Metric Tool                                                                           | e-Surgeon (PowerAPI)                              | BitWatts                               | PowerJoular                                    | Joular Core                                    | e-Surgeon (eJalen)                                                 | JoularJX                                                                                                                                 | IgProf (Energy module) | JRAPL                                   |    |
|---------------------------------|-----------------------------------------------------------------------------|-----------------------------------|-----------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------|------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------|----|
| Granularity                     | Virtual Machine-Level                                                       |                                   |                                   | X                                          | X                                                                                                             | X                                                                                                                | no                                                                                          | no                                                | no                                     | X                                              | X                                              | no                                                                 | no                                                                                                                                       | X                      | no                                      |    |
|                                 | Pod-Level                                                                   |                                   |                                   | X                                          | X                                                                                                             | X                                                                                                                | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 | Container-Level                                                             |                                   |                                   | X                                          | X                                                                                                             | X                                                                                                                | X                                                                                           | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 | Application-Level                                                           |                                   |                                   | X                                          | X                                                                                                             | X                                                                                                                | no                                                                                          | no                                                | no                                     | X                                              | X                                              | no                                                                 | no                                                                                                                                       | X                      | no                                      |    |
|                                 | Process-Level                                                               |                                   |                                   | X                                          | X                                                                                                             | X                                                                                                                | no                                                                                          | X                                                 | X                                      | X                                              | X                                              | no                                                                 | no                                                                                                                                       | X                      | no                                      |    |
|                                 | Thread-Level                                                                |                                   |                                   | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | X                                                                  | X                                                                                                                                        | X                      | no                                      |    |
|                                 | Code Section-Level                                                          |                                   |                                   | no                                         | no                                                                                                            | no                                                                                                               | X(SCI functionality)                                                                        | no                                                | no                                     | no                                             | no                                             | X                                                                  | X                                                                                                                                        | X                      | X                                       |    |
| Measurement Sources             | CPU                                                                         | Hardware-Level Energy Estimation  | External Power Meter              | no                                         | no                                                                                                            | no                                                                                                               | X(Compatible with certain Metric Providers)                                                 | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             |                                   | Hardware-Exposed Energy Interface | no                                         | X(RAPL)                                                                                                       | X(RAPL)                                                                                                          | X(RAPL)                                                                                     | no                                                | no                                     | X(RAPL through powercap)                       | X(RAPL (linux/windows or powerMetrics (Mac)))  | no                                                                 | X(RAPL through powercap)                                                                                                                 | X(RAPL through PAPI)   | X(RAPL through linux MSR kernel module) |    |
|                                 |                                                                             | Model-based                       | no                                | XRegression-based models for VM deployment | XRegression-based models for VM deployment                                                                    | X1) XGBoost model (TDP, Machine Specs and CPUs)<br>2) Analytical Model(TDP, #CPUs and CPU %)                     | XLinear Regression Model (HPC: "Instructions", "cache references", "cache misses")          | XHPC vector (cycles, instructions, cache misses)  | no                                     | no                                             | X(TDP) Through PowerAPI                        | no                                                                 | no                                                                                                                                       | no                     |                                         |    |
|                                 |                                                                             |                                   | Software Activity Share Proxy     | no                                         | OS-provided PID CPU %<br>Cgroups Stat                                                                         | PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing                               | OS-provided PID CPU %<br>Cgroups Stat<br>Code Special Print (Code Section)                  | OS-provided PID CPU %                             | OS-provided PID CPU %                  | OS-provided PID CPU %                          | OS-provided PID CPU %                          | 1) Time Markers : JVM byte code instrumentation / 2) Call stack    | 1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Call Stack                                                      | Call stack             | Time Markers                            |    |
|                                 | DRAM                                                                        | Hardware-Level Energy Estimation  | External Power Meter              | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             |                                   | Hardware-Exposed Energy Interface | no                                         | X(RAPL)                                                                                                       | X(RAPL)                                                                                                          | X(RAPL)                                                                                     | no                                                | no                                     | X(RAPL)                                        | X(RAPL)                                        | no                                                                 | X(RAPL) Through Power Joular                                                                                                             | X(RAPL)                | X(RAPL through linux MSR kernel module) |    |
|                                 |                                                                             | Model-based                       | no                                | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             |                                   | Software Activity Share Proxy     | no                                         | OS-provided PID CPU %<br>Cgroups Stat                                                                         | PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing                               | OS-provided PID CPU %<br>Cgroups Stat (Container)<br>Code Special Print (Code Section)      | no                                                | no                                     | PID CPU %                                      | PID CPU %                                      | no                                                                 | 1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace                      | Call stack sampling    | Time Markers                            |    |
|                                 | GPU                                                                         | Hardware-Level Energy Estimation  | External Power Meter              | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      | no |
|                                 |                                                                             |                                   | Hardware-Exposed Energy Interface | no                                         | X(NVML)                                                                                                       | X(NVML)                                                                                                          | X(NVML)                                                                                     | no                                                | no                                     | X(NVIDIA-SMI, AMD-SMI) (computers and servers) | X(NVIDIA-SMI, AMD-SMI) (computers and servers) | no                                                                 | X(NVIDIA SMI) (computers and servers through Power Joular)                                                                               | no                     | no                                      |    |
|                                 |                                                                             | Model-based                       | no                                | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      | no |
|                                 |                                                                             |                                   | Software Activity Share Proxy     | no                                         | NVMLM-provided (i.e. driver) per-process SM utilization                                                       | NVMLM-provided (i.e. driver) per-process SM utilization                                                          | OS-provided PID CPU %<br>Cgroups Stat<br>Code Special Print (Code Section)                  | no                                                | no                                     | PID GPU % (from driver API)                    | PID GPU % (from driver API)                    | no                                                                 | 1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace                      | no                     | no                                      |    |
| Network (NIC)                   | Hardware-Level Energy Estimation                                            | External Power Meter              | no                                | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             | Hardware-Exposed Energy Interface | no                                | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             | Model-based                       | no                                | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             |                                   | Software Activity Share Proxy     | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | X<br>OS provided per-process (PID)Network I/O     | no                                     | no                                             | no                                             | X(Through PowerAPI)<br>getInputStream()<br>getOutputStream() calls | no                                                                                                                                       | no                     | no                                      | no |
|                                 | System                                                                      | Hardware-Level Energy Estimation  | External Power Meter              | no                                         | X(IPMI, ACPI)                                                                                                 | X(IPMI, ACPI)                                                                                                    | X                                                                                           | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             |                                   | Hardware-Exposed Energy Interface | X(RAPL)                                    | no                                                                                                            | no                                                                                                               | X(RAPL PSYS Domain)                                                                         | no                                                | no                                     | no                                             | X(RAPL)                                        | no                                                                 | no                                                                                                                                       | X(RAPL through PAPI)   | X(RAPL through linux MSR kernel module) |    |
|                                 |                                                                             | Model-based                       | no                                | Regression-based model                     | Regression-based trained power model                                                                          | X1) XGBoost model (TDP, Machine Specs and CPUs)<br>2) Analytical Model(TDP, #CPUs and CPU %)                     | X(TDP)                                                                                      | X HPC (cycles, instructions, cache misses) vector | polynomial regression For Raspberry Pi | polynomial regression For Raspberry Pi         | no                                             | polynomial regression For Raspberry Pi                             | no                                                                                                                                       | no                     |                                         |    |
|                                 |                                                                             |                                   | Software Activity Share Proxy     | OS-provided PID CPU %                      | OS-provided PID CPU %<br>Cgroups Stat                                                                         | PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing                               | OS-provided PID CPU %<br>Cgroups Stat (container)<br>Code Special Print (Code Section)      | OS-provided PID CPU %                             | OS-provided PID CPU %                  | OS-provided PID CPU %                          | OS-provided PID CPU %                          | no                                                                 | 1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace                      | Call stack sampling    | Time Markers                            |    |
| Allocation Model                | Activity Share Estimation                                                   | Directly proportional to          | X                                 | X                                          | no                                                                                                            | X                                                                                                                | X                                                                                           | X                                                 | X                                      | X                                              | X (For NIC)                                    | X(for threads)                                                     | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             | Proxy Reconstruction              | no                                | no                                         | X (from HPC vector (cycles, instructions, cache misses))                                                      | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 |                                                                             | Statistical Sampling              | no                                | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | X(for CPU : call-stack sampling version)       | X                                                                  | X(1) Allocate all RAPL system energy to process it is attached to 2) Statistical Sampling (stack trace) to allocate energy to functions) | no                     | no                                      |    |
|                                 |                                                                             |                                   | Execution Time Markers            | no                                         | no                                                                                                            | no                                                                                                               | X(SCI functionality)                                                                        | no                                                | no                                     | no                                             | no                                             | X (for CPU : bytecode instrumentation version)                     | no                                                                                                                                       | no                     | X                                       | no |
|                                 | Baseline Modelling and Calibration                                          |                                   |                                   | no                                         | Modeled and allocated proportionnaly to processes size                                                        | Modeled and allocated proportionnaly to processes size                                                           | Modeled and allocated to container according to OS-provided PID CPU %                       | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
| Virtualised environment support | Pretrained Model                                                            |                                   |                                   | no                                         | Regression model (# CPU and OS-provided Resource Utilisation metrics as input)                                | Regression model (# CPU and OS-provided Resource Utilisation metrics as input)                                   | 1) XGBoost model (TDP, Machine Specs and CPU%)<br>2) Analytical Model(TDP, #CPUs and CPU %) | no                                                | no                                     | no                                             | no                                             | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 | Two instances : host exposing VM-level metrics to guest through shared file |                                   |                                   | X                                          | ((CNCF 2023) mentions a planned "passthrough" mode for VM deployment which expose VM energy in a shared file. | no ((CNCF 2023) mentions a planned "passthrough" mode for VM deployment which expose VM energy in a shared file. | no                                                                                          | no                                                | no                                     | X                                              | X                                              | no                                                                 | no                                                                                                                                       | no                     | no                                      |    |
|                                 | Delegated to another tool                                                   |                                   |                                   | no                                         | no                                                                                                            | no                                                                                                               | no                                                                                          | no                                                | no                                     | no                                             | no                                             | no                                                                 | Uses PowerJoular                                                                                                                         | no                     | no                                      |    |